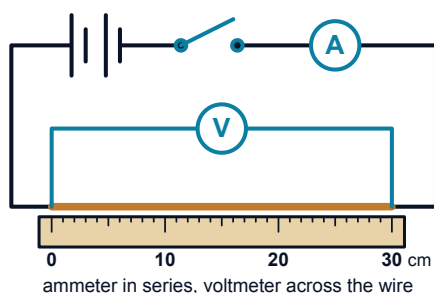


# Resistance and the length of a wire

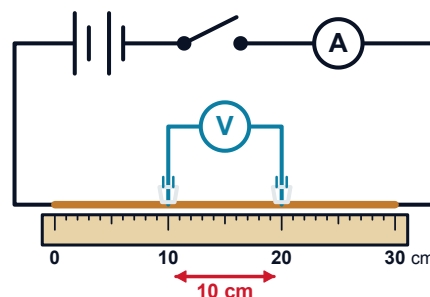
Electricity  
Paper 1 · Combined and Separate  
6 steps

A student investigates how the length of a wire affects its resistance.

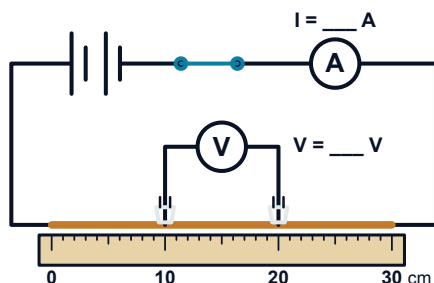
- 1** Connect the wire in a circuit, with an ammeter in series and a voltmeter across the wire.



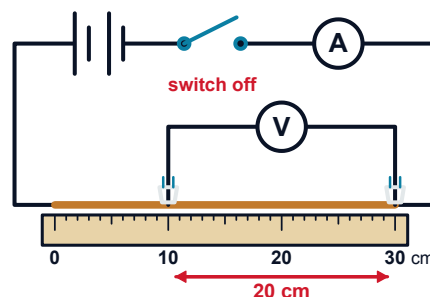
- 2** Attach the crocodile clips 10 cm apart on the wire.



- 3** Switch on and record the current and the potential difference.



- 4** Switch off to let the wire cool, then move the clips to 20 cm apart.



- 5** Repeat the readings for a range of lengths.

| length (cm) | I (A) | V (V) |
|-------------|-------|-------|
| 5           |       |       |
| 10          |       |       |
| 15          |       |       |
| 20          |       |       |
| 25          |       |       |
| 30          |       |       |

- 6** Calculate the resistance for each length and plot resistance against length.

$$R = V \div I$$

(for each length)

